

Dead Signal

Localization Plan

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Purpose. How *Dead Signal* becomes translatable and ships in more than one language. The plan is front-loaded on one hard dependency — **string externalization** — because nothing else in localization can begin until the text is extracted from the game logic. Priority in the [PRD](#) is **P2** (a polished-launch/fast-follow goal, not ship-blocking).

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1. Readiness assessment

Current state: English-only, fully hardcoded. Every player-facing string is authored inline in `src/DeadSignal.jsx` — inside the module-scope data pools (`INTRO_LINES`, `PATH_BEATS`, `EXPLORE_BEATS`, `STATE_LINES`, `HAVEN_*`, `ENCOUNTERS`, `BOARD_*`, the finale/ending arrays, UI labels) and in the render code. There is **no i18n scaffolding** — no string table, no locale switch, no message-format layer.

Scale. The authored surface is large for a text game: on the order of **several hundred string literals** (the data pools alone carry hundreds of `msgs/choices/text/name/note` entries), plus UI chrome (menus, HUD registers, Options, Case File labels). This is the product — it is text-dense by nature.

Implication. Localization is **not a translation task first; it is a refactor first**. Until strings are externalized, there is nothing for a translator to touch. This document treats externalization as milestone zero.

2. The critical path: string externalization

Extract every player-facing string out of `DeadSignal.jsx` into a **locale catalog**, and have the game read from it by key.

Approach (recommended):

1. **Inventory & tag.** Sweep the data pools + render layer; mark each string as *content* (story beats, Ellie/narrator lines) vs *chrome* (menus, HUD, labels). Content and chrome localize differently (content is voiced; chrome is terse and length-critical).
2. **Key every string.** Give each a stable key (\$3). Story beats key by their existing structure (e.g. `beat.hospital.leg1.enc2.msg[0]`), chrome by function (`ui.menu.start`).
3. **Externalize to en catalog.** Move the strings into a base-language catalog file(s); replace inline literals with key lookups via a tiny `t(key)` helper. **The game must play identically after this step** (English from the catalog) — that's the checkpoint (mirrors the “never break the build” rule from `PLAN.md`).
4. **Only then** are other locales just additional catalogs.

Determinism is preserved. Catalogs are static local assets bundled at build — no network, no runtime fetch (respects PRD NFR-1). Locale is chosen from device/browser language with a manual override in Options.

Expansion v2 · build: PLANNED — the Phase-3 surface roughly doubles. The Expansion v2 narrative pass (`STORY.md` \$3/\$5/\$8 + Changelog — Expansion v2) adds a large block of *new translatable content*: **~14 Echo fragments** (2–4 lines each, the 7-face cast), **~30 new Phase-3 node onEnter/atmosphere strings** (region sub-stories; Phase 3 text roughly **doubles**, ~35 → ~65 nodes), the “**unchosen**” strings (City Hall selection/petitions/dissent) with the **evolving**

Case File question (why only 143? → who did you leave out — and why?), the **prologue second-act** strings (route midpoints, the shelter scene, extra Ellie cracks), and one new Case File category label **ECHOES**. **Critical-path implication:** externalization (M-LOC0) is *still* the gate, and it must land **before** this content is authored into the build. If the ~2× of new Phase-3 text is written back into inline literals in DeadSignal.jsx, the extraction debt compounds — the milestone- zero refactor grows against a moving target. Externalize first, then let the Expansion v2 content land as keyed strings from day one. (No code exists for any of this yet; all of it is design-only.)

3. String architecture

- **Format.** A flat or lightly-nested key→string map per locale (JSON is the pragmatic choice; it diffs well and every TMS speaks it). One base en catalog is the source of truth translators work from.
 - **Key schema.** <domain>.<area>.<id> — e.g. story.haven.final.call, ui.hud.battery, casefile.question.kim, ending.accept.line3. Keys are English-descriptive and **never change meaning** once shipped (changing a string = new key or a versioned key, so stale translations are detectable).
 - **Placeholders & format.** Use a message-format standard (**ICU MessageFormat**) for any interpolation (resource counts, [+N Food], 143) and for **plurals** — plural rules differ per language (e.g. Slavic languages have 3–4 plural forms; battery “%d%%” is fine but any “N items” needs ICU). Keep placeholders named, not positional.
 - **Registers as metadata.** Tag each content string with its **voice** (ellie / narrator / player-choice / hud) so translators keep the register (§5) and so a per-locale style guide can target them. **Expansion v2 (build: PLANNED)** adds an **echo** voice tag for the recovered fragments of the 142 minds — a distinct, terse, mid-thought register that must not be flattened into narrator (§5.1).
 - **What NOT to externalize.** Purely numeric/· continuation tokens, internal ids, the 143 number itself (it is a number, not a word — §5), and debug strings.
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4. Target languages & tiers

A proposed rollout. Final selection is a business decision (PRD open question) driven by the target markets and the distribution channel.

Tier	Languages	Rationale
Pilot	1 language (suggest Spanish, es-419 or French, fr)	Latin script (no new font), large audience, validates the whole pipeline cheaply
Tier 1	FR, DE, ES-419, PT-BR, IT	Latin-script, high-ROI Western markets; all covered by the current font

Tier 2	JA, ZH-Hans, KO, RU	High-value but each adds script/font work (CJK, Cyrillic) — see §6
Tier 3+	ZH-Hant, PL, TR, others	Demand-driven expansion

Sequencing logic: prove the pipeline on one Latin language, ship Tier 1 (no font changes), then take on Tier 2 where each language carries extra typographic cost. Interactive-fiction audiences skew toward EN/JA/ZH/DE/FR/RU, so Tier 2 is where much of the remaining reach is — plan for it, don’t front-load it.

5. Culturalization & voice

The traps that a straight translation would break. These need **per-locale creative decisions**, not literal translation.

- **The 143 motif — mostly safe, one nuance.** In-game, **143 is the resident count** (the status board, the bunks, the player’s held slot). As a *number* it is universal and must stay 143 everywhere. The *cultural resonance* — 143 as pager/text shorthand for “I love you” (letter counts 1-4-3) — is an **English-culture Easter egg**, not load-bearing. Do not try to localize it; just ensure no translated line leans on the “love” reading. Document it so translators understand why the number is fixed. **Expansion v2 note (build: PLANNED):** the motif now also underpins **Echo / roster content** — the 143 is a *count of people* (bunk 143 = Kim’s empty refused bunk, the roster the architect signed, the evolving why only 143? question). Keep it fixed as a number there too; the watch-item is unchanged but its footprint grows.
- **KIM → ELLIE contact-name flip.** Proper names carry across languages; keep **Kim** and **Ellie** as-is (or apply standard transliteration for non-Latin scripts — e.g. キム／エリ — in JA). The *beat* (the reader noticing the name change) works in any language — protect it by keeping the flip visually obvious in every locale. **Expansion v2 note (build: PLANNED):** the flip’s meaning now extends into Echo/roster content — Kim is deliberately **not** an Echo (the Signal Core “absence beat,” STORY.md §3), while Ellie is the connected voice; the contrast the flip sets up must stay legible wherever both names appear across Phase 3.
- **Ellie’s lowercase, terse register — the hard one.** Ellie speaks in **all-lowercase** as a characterization (diminished, tired, intimate). **Case is an English/Latin device** — CJK scripts have no case, so “lowercase” conveys nothing there. Each such language needs an **equivalent register** (e.g. in JA: plain/casual form, minimal particles, no honorifics; in general: clipped sentences, lowercase where the script supports it, informal register). This is a **style-guide deliverable per language**, agreed with the translator, not a mechanical rule.
- **The narrator’s centered italic, sparse voice.** Keep the terseness and present-tense immediacy; italics render fine cross-script but carry less weight in CJK — pair with the existing centering.

- **Spoiler discipline is a translation constraint.** Translators must preserve the *cracks-not-answers* rule (STORY.md §4): a line that merely implies must not become a line that states. Ship translators the relevant STORY.md context under NDA so they translate the subtext correctly.

5.1 Expansion v2 culturalization watch-items — build: PLANNED

New traps introduced by the Expansion v2 narrative pass (STORY.md §3/§5/§8). Design-only; no code exists yet. Fold these into the per-language style guide alongside the items above.

- **The child Echo (Theo) — register + content-rating sensitivity.** Theo is a child (~8); his Echo is a child processing upload as “*a nap on the bus*” (STORY.md §3). Two demands per locale: (1) an **age-appropriate child register** — the diction of an ~8-year-old in that language, which is not a mechanical translation of the English; and (2) **content-rating sensitivity** — a child facing death is handled differently across ratings boards/markets (some locales are stricter about implied harm to minors). Flag Theo’s fragments for **per-market classification review** before ship; the source restraint (short, oblique, never staged for shock — §4) is what keeps it shippable, and translation must *preserve* that restraint rather than sharpen it.
- **The terse Echo register — restraint must survive translation.** Echoes are **2–4 lines, found, mid-thought — never a scene, never a confession** (STORY.md §4: “Echoes crack, they don’t confess”). The failure mode in translation is **melodrama**: a language’s natural instinct to soften, explain, or emotionally underline will turn a fragment into a speech. **Restraint is canon** — the child (Theo) and the regretter (Priya) are the hardest lines in the game *because they are the shortest*. Give translators an explicit brief: match the clipped length, keep the ambiguity, do not add connective/emotional scaffolding. Treat this as a **new voice tag** (echo, §3) with its own style-guide entry, distinct from narrator.
- **The “unchosen / who did you leave out” moral framing.** The Expansion v2 spine (STORY.md §2, the unchosen) turns on a moral accusation — *you signed who wasn’t saved* — carried by City Hall’s selection/petitions/dissent, Marcus’s Echo, and the evolving question why only 143? → who did you leave out — and why?. The framing is **accusatory-but-restrained**, second-person, and must not tip into either sermon or melodrama in any locale. The pivot from **quantity** (“only 143”) to **culpability** (“who did *you* leave out”) is the load-bearing shift — protect the second-person “you” and the shift itself; some languages will need a deliberate choice of address/formality to keep the accusation personal (see the Ellie register note on informality above).
- **Character names — consistency across locales (build: PLANNED).** Expansion v2 adds a named cast that recurs across many strings (Echoes, roster, City Hall, Case File PEOPLE). **Lock a single rendering per name per locale** to avoid drift: **Theo · Rosa · Walt · Priya · Marcus · Dr. Lena Sorkin · June · Kim · Ellie**. Latin-script locales keep them as-is; non-Latin scripts apply a **fixed, agreed transliteration** (as with キム／エリ — for Kim/Ellie) recorded in a per-locale name glossary so the *same* character reads identically everywhere. Watch **Dr. Lena Sorkin** specifically: keep the honorific/title

handling consistent (title order and abbreviation of “Dr.” differ per language), and keep first/last-name usage matching how the source deploys it.

6. Typography & script coverage

- **Current font: IBM Plex Mono** (Google Fonts, weights 300/400), fallback 'Courier New', monospace. IBM Plex Mono covers **Latin, Greek, and Cyrillic** — so **Tier 1 + Russian** are typographically ready.
- **CJK is the gap.** IBM Plex Mono has **no CJK glyphs**. Japanese/Chinese/Korean need a different face — IBM ships **IBM Plex Sans JP/KR/TC/SC** (keeps the brand family) or fall back to **Noto Sans Mono CJK**. Note CJK “monospace” is really *duospace* (full-width glyphs) — the terminal grid look will shift; validate the aesthetic per script (Art Bible §4 is Latin-specific).
- **Font loading & offline.** The current @import pulls the font from Google Fonts at runtime. For determinism/offline (PRD NFR-1) and for large CJK faces, **self-host and subset** the fonts and bundle them; lazy-load the CJK face only for CJK locales to ke